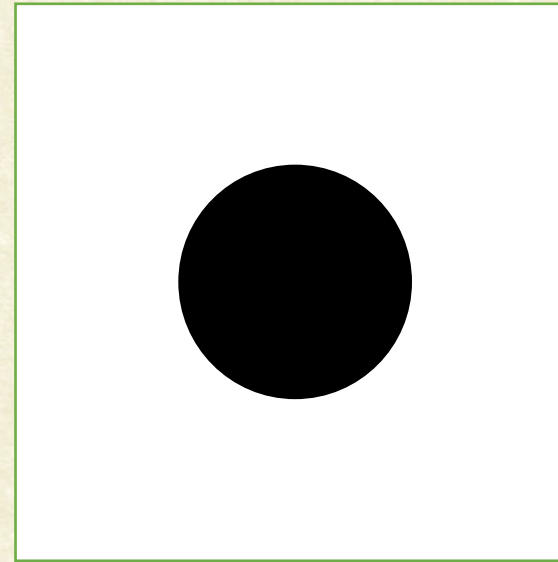
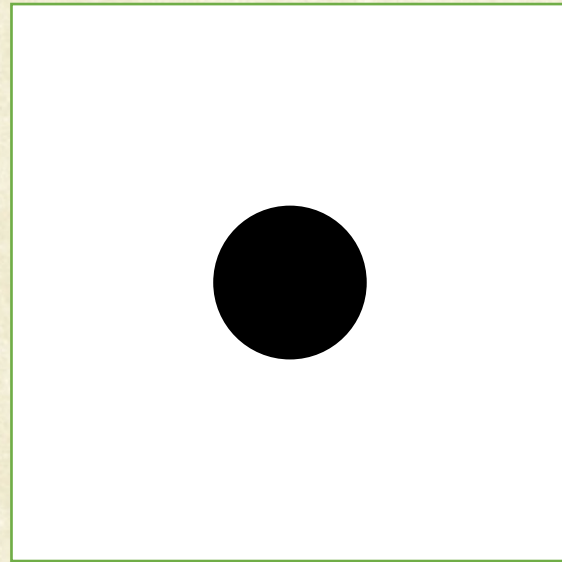




CS7.404: Digital Image Processing

Monsoon 2025: Morphological Image Processing 2



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Prove the Following:

Morphological Opening:

1. $A \circ B$ is a subset of A
2. If C is a subset of D , then $C \circ B$ is a subset of $D \circ B$
3. $(A \circ B) \circ B = A \circ B$

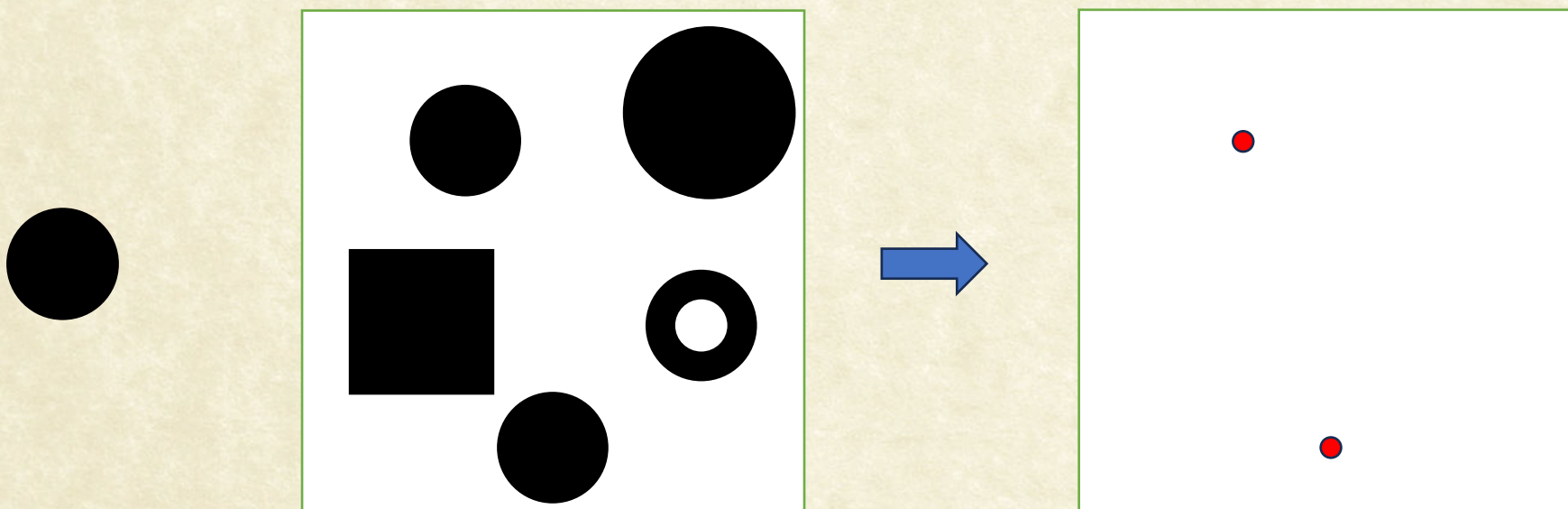
Morphological Closing:

1. A is a subset of $A \cdot B$
2. If C is a subset of D , then $C \cdot B$ is a subset of $D \cdot B$
3. $(A \cdot B) \cdot B = A \cdot B$



CS7.404: Digital Image Processing

Monsoon 2025: Hit or Miss Transform (HMT)



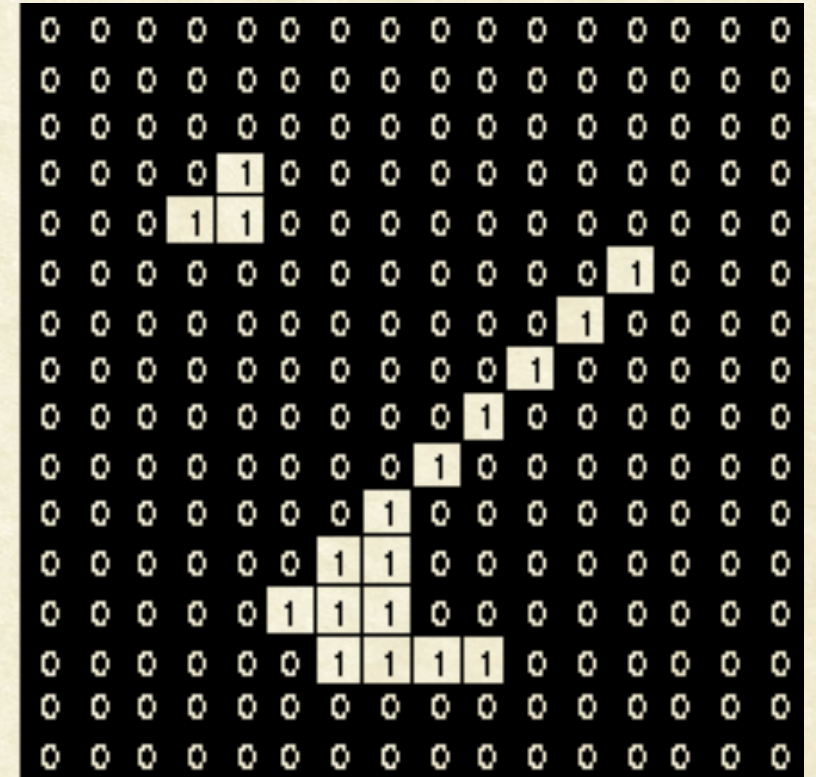
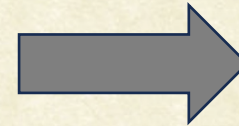
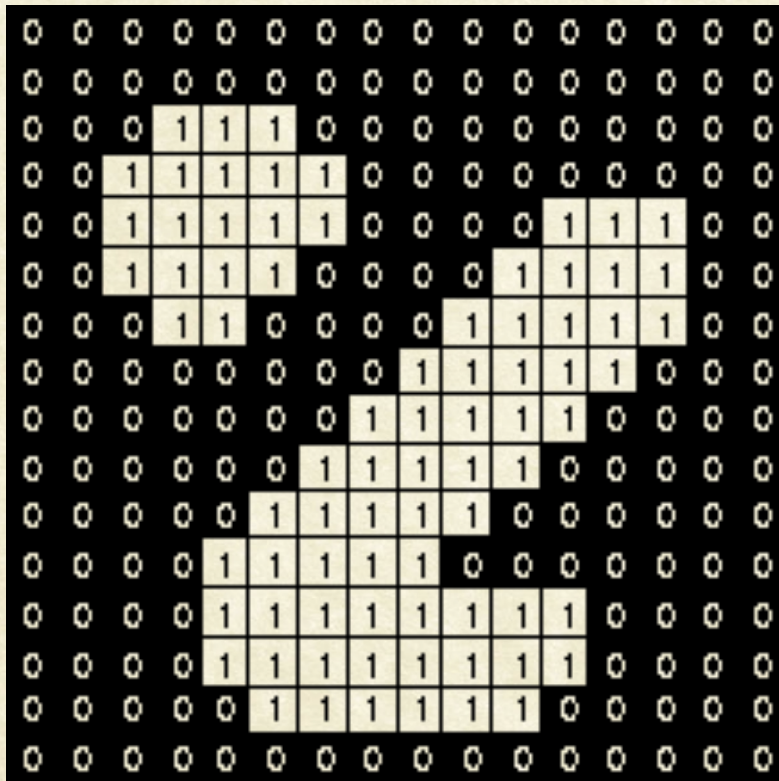
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Recap: Erosion

1	1	1
1	1	1
1	1	1

- Simple application of **pattern matching**





HMT: Finding Exact Matches

- Goal:
 - Detect objects of a particular shape in the image (exact matches)
- Approach:
 - Find locations in the image where:
 1. The shape (B_1) fits inside the foreground
 2. The boundary of the shape (B_2) fits in the background
 - Take the intersection of the two to find object locations

- Definition:

$$\begin{aligned} I \circledast B_{1,2} &= \{z | (B_1)_z \subseteq A \text{ and } (B_2)_z \subseteq A^c\} \\ &= (A \ominus B_1) \cap (A^c \ominus B_2) \end{aligned}$$



HMT Example

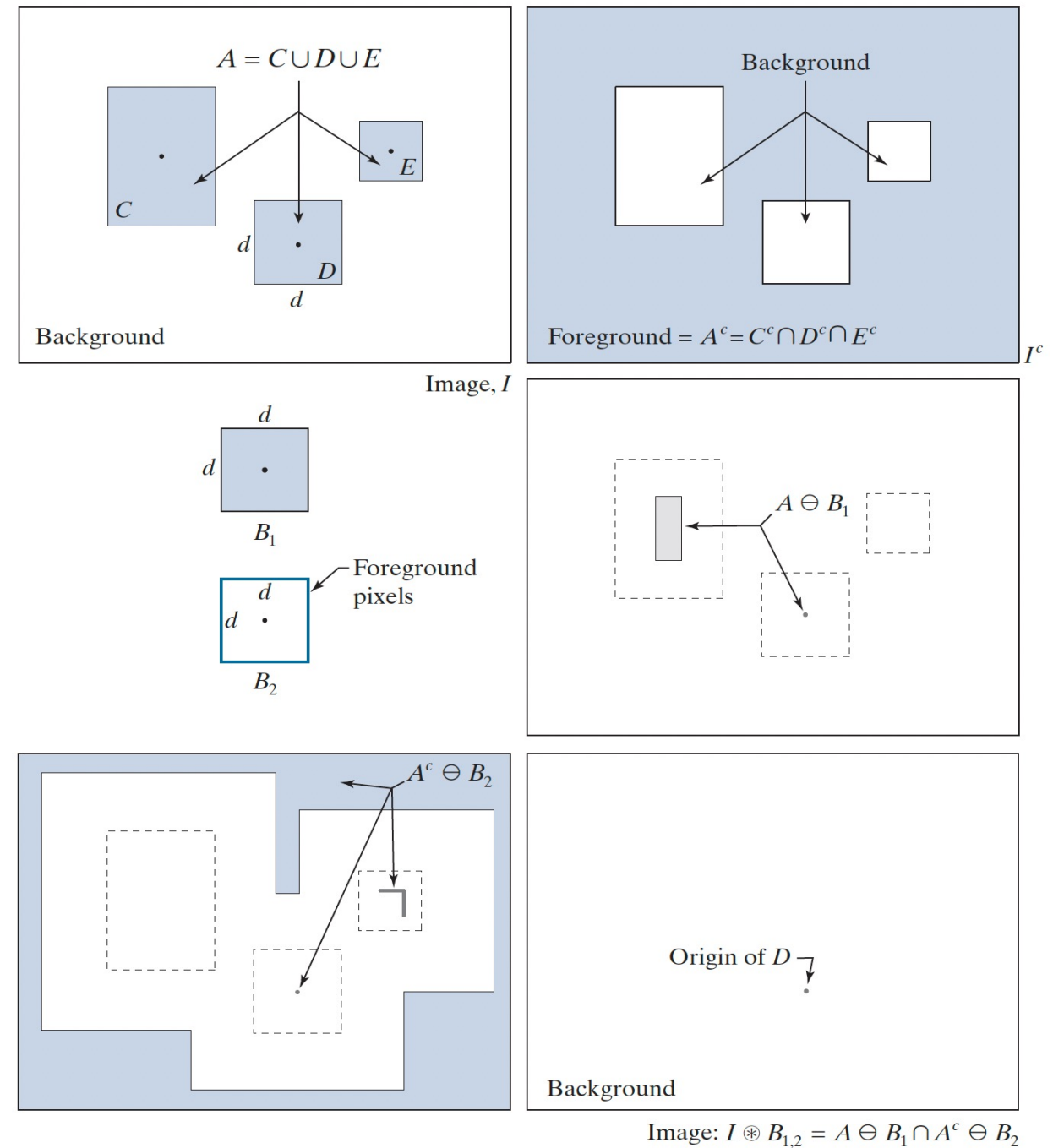
Row1: Original Image and its complement

Row 2:

- The structuring elements
- Hits with B_1 .

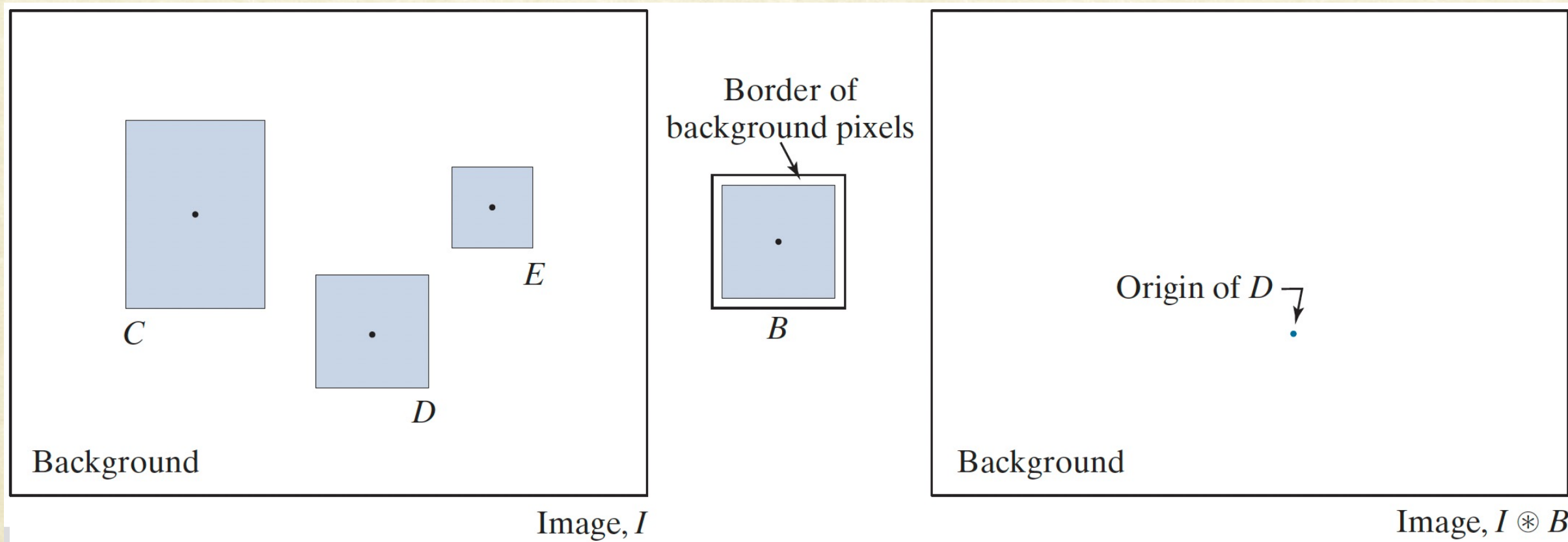
• Row 3:

- Hits with B_2 .
- Intersection of the hits





HMT with single structuring element



- We match backgrounds as well

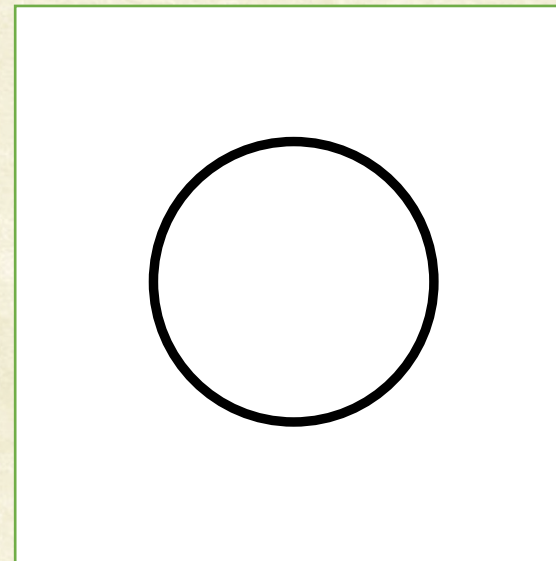
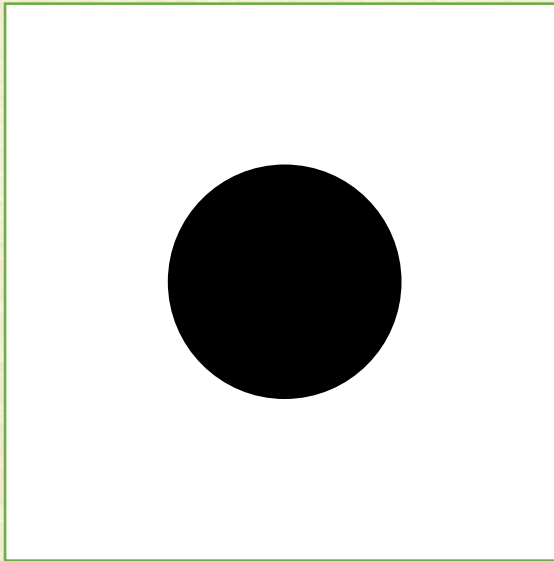


Questions?



CS7.404: Digital Image Processing

Monsoon 2025: Boundary Detection



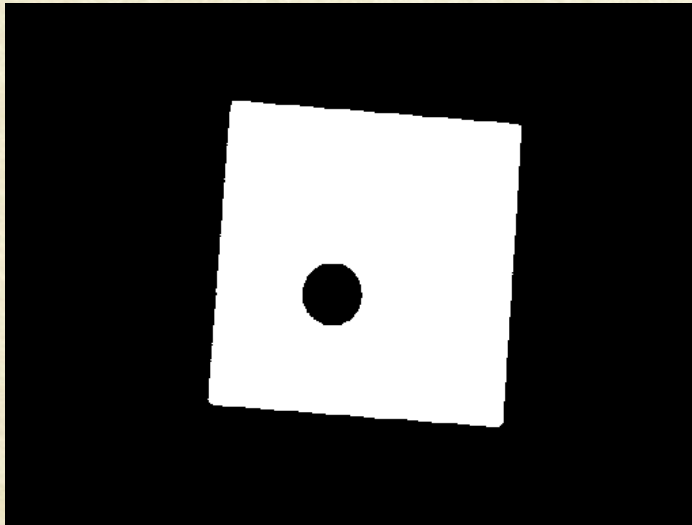
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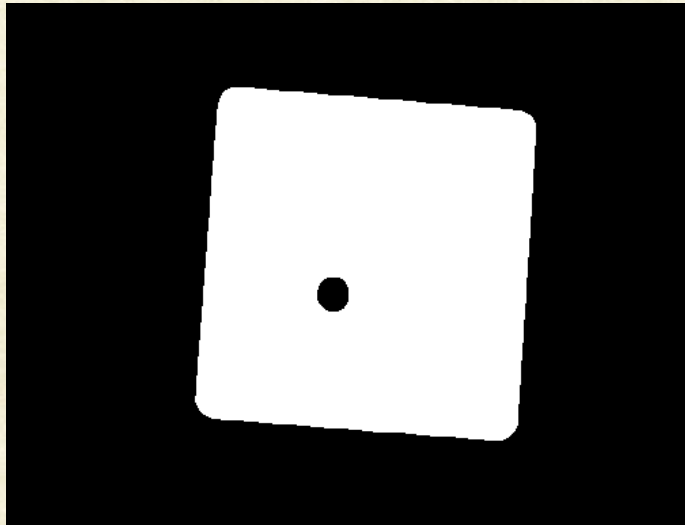
Boundary Detection

1. Dilate input image
2. Subtract input image from dilated image
3. Boundaries remain!

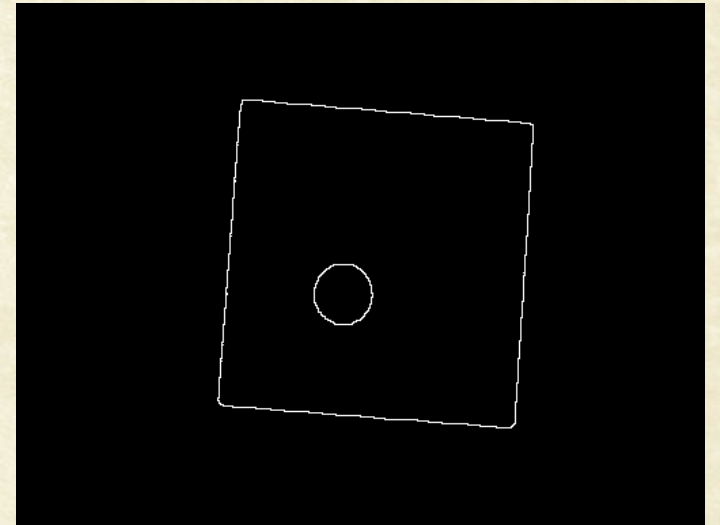
A



$(A \oplus B)$



$\beta(A) = (A \oplus B) - A$





Erosion can also be used to find Boundary

A



$(A \ominus B)$

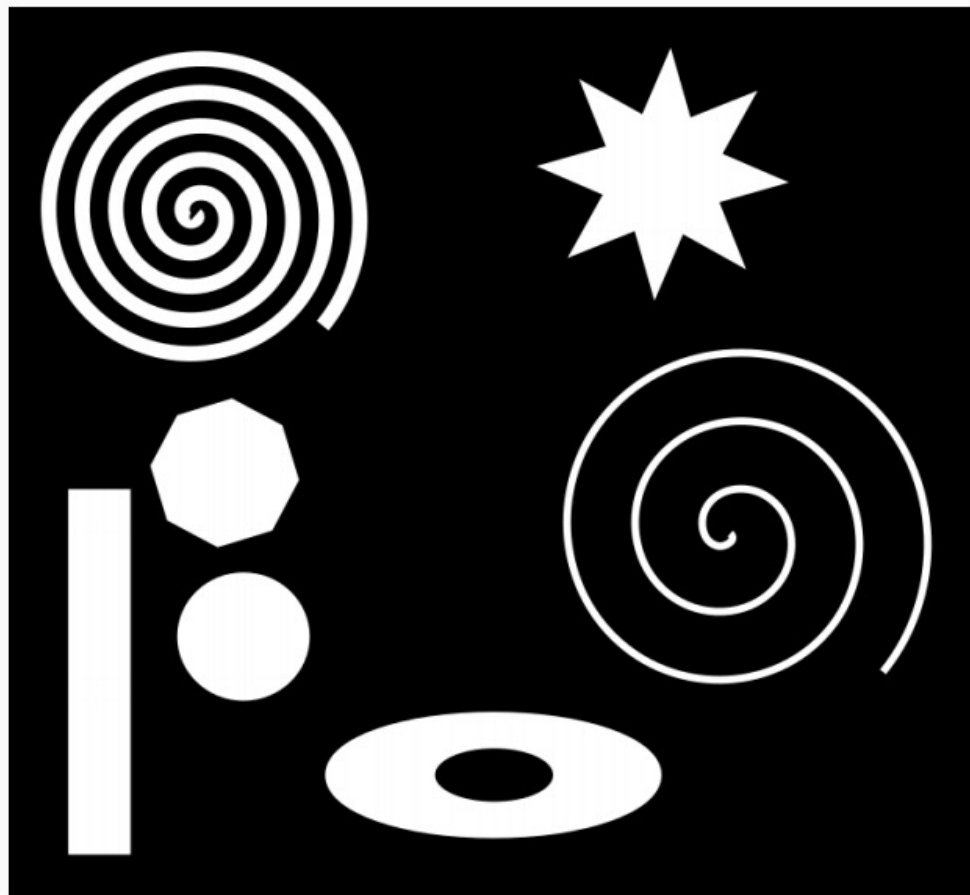


$\beta(A) = A - (A \ominus B)$





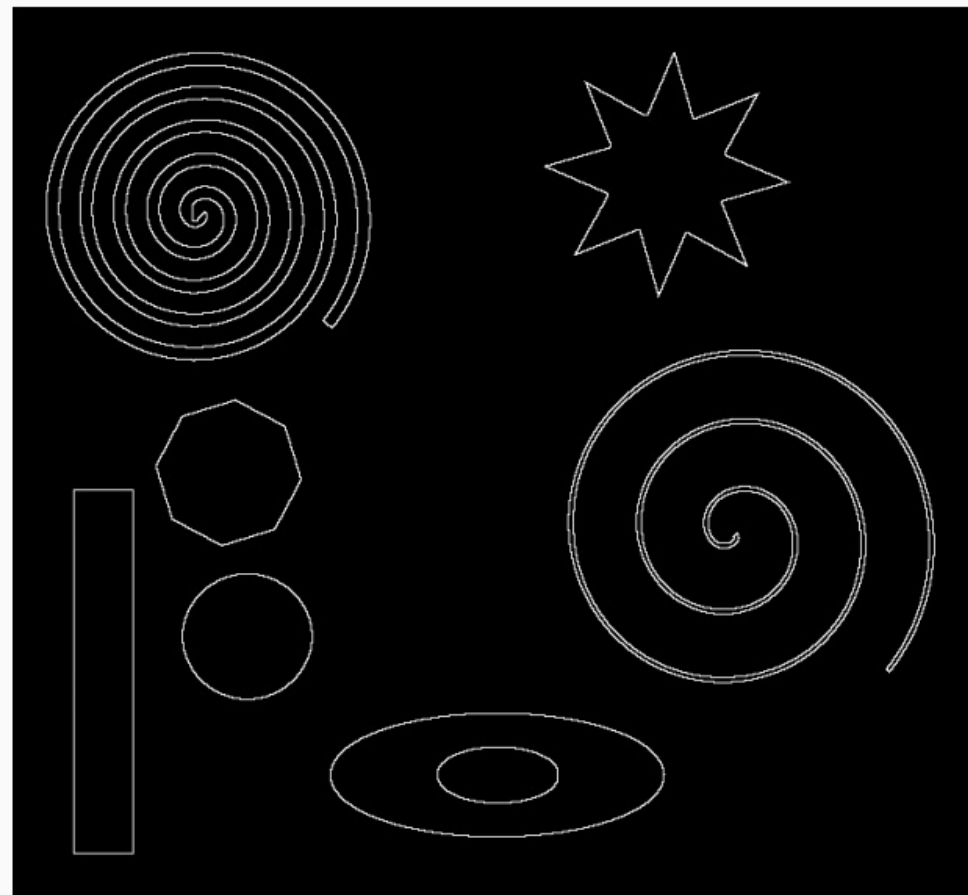
Boundary Detection: Effect of STREL



(a) f

1	1	1
1	1	1
1	1	1

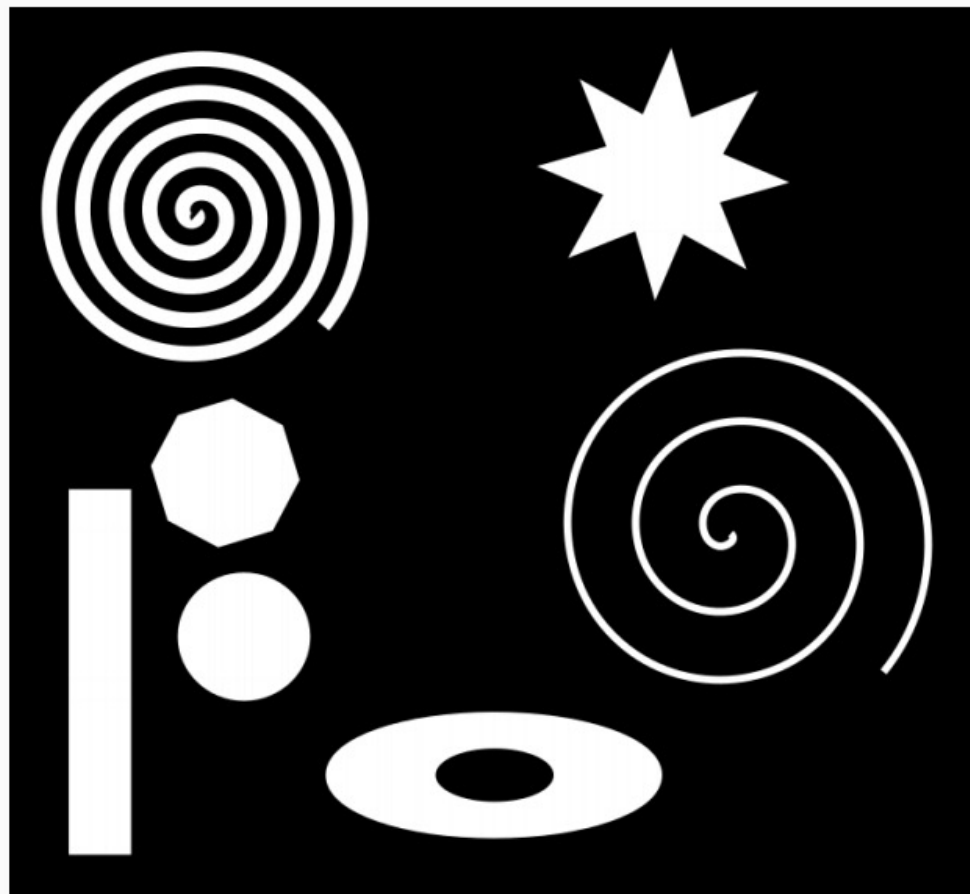
(b) s



(c) $f - (f \ominus s)$



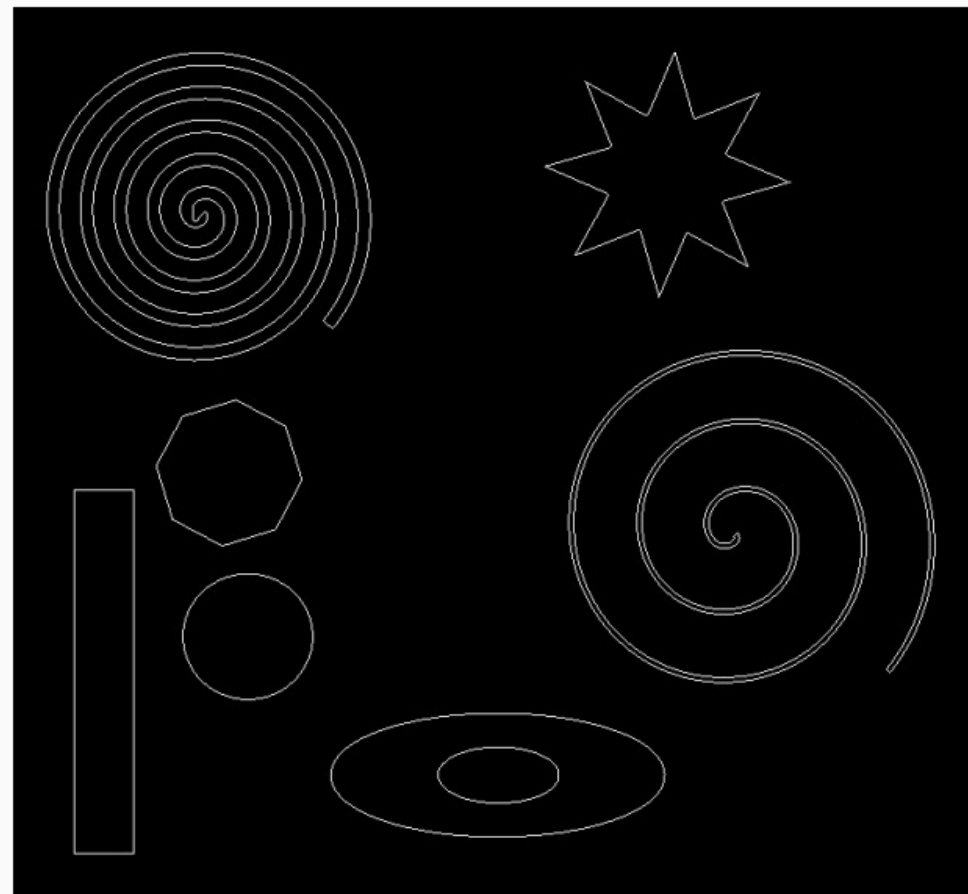
Boundary Detection: Effect of STREL



(a) f

0	1	0
1	1	1
0	1	0

(b) s



(c) $f - (f \ominus s)$

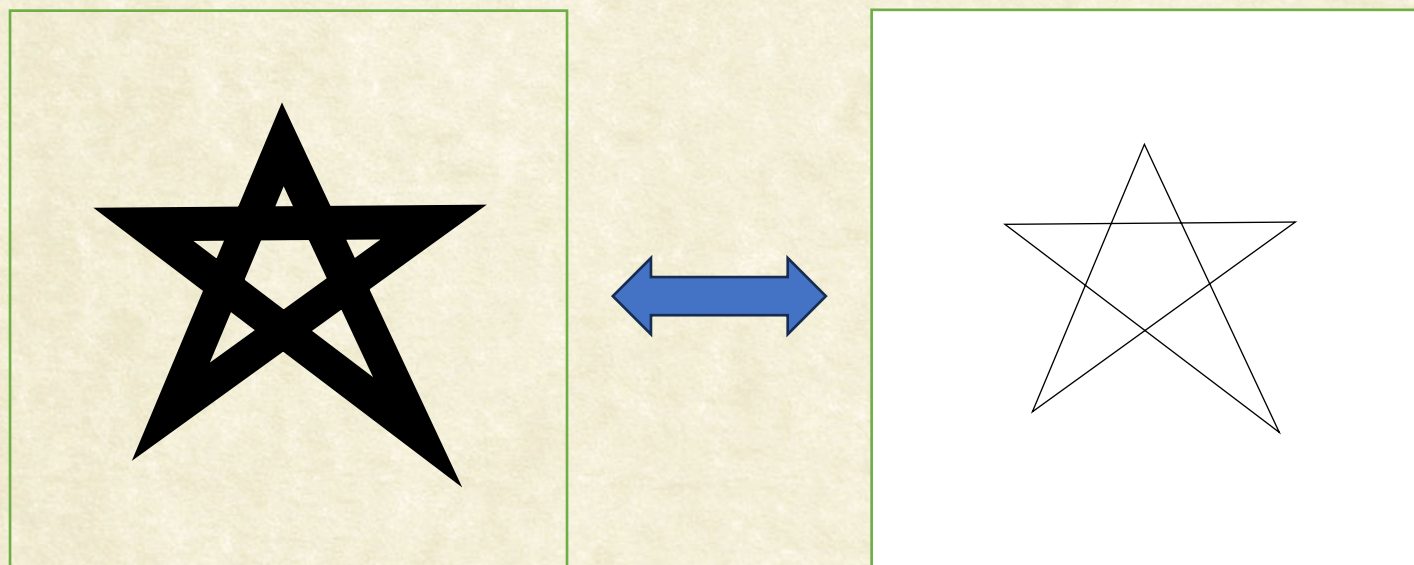


Questions?



CS7.404: Digital Image Processing

Monsoon 2025: Thinning and Thickening



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Thinning

- In terms of HMT:

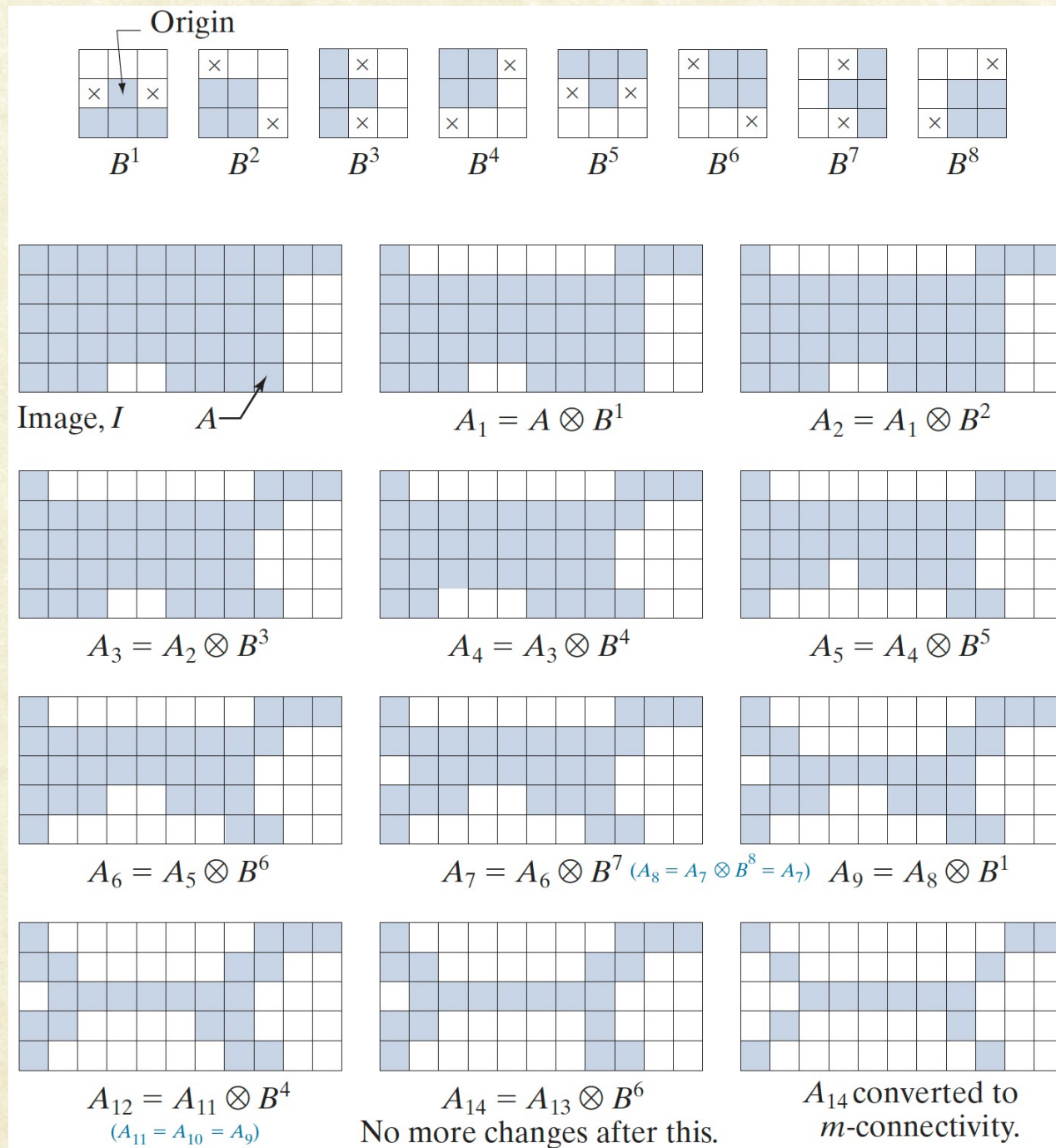
$$A \otimes B = A - (A * B)$$

$$= A \cap (A * B)^c$$

- Repeat with a set of specific structuring elements:

$$A \otimes \{B\} = ((...((A \otimes B^1) \otimes B^2) ...) \otimes B^n)$$

until no more changes



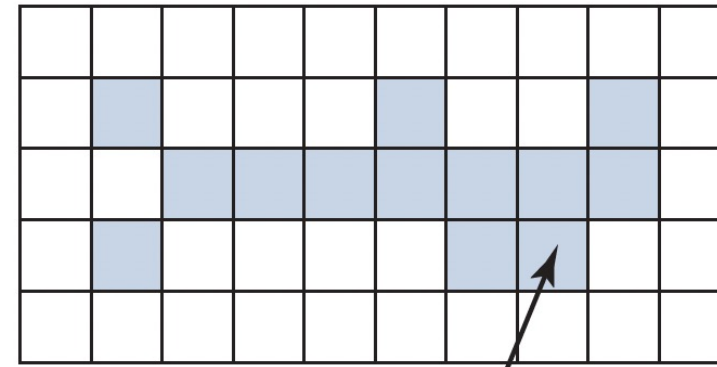


Thickening

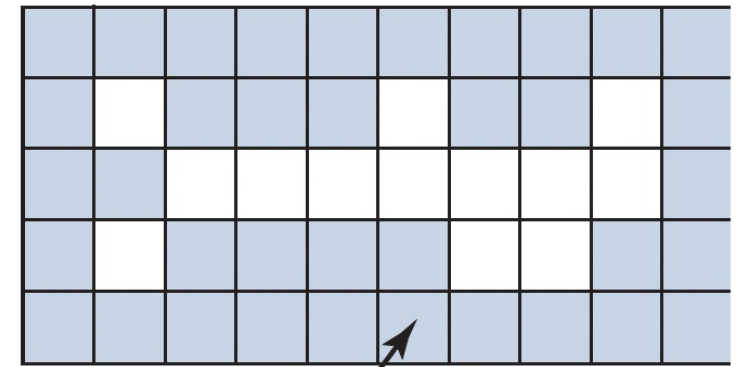
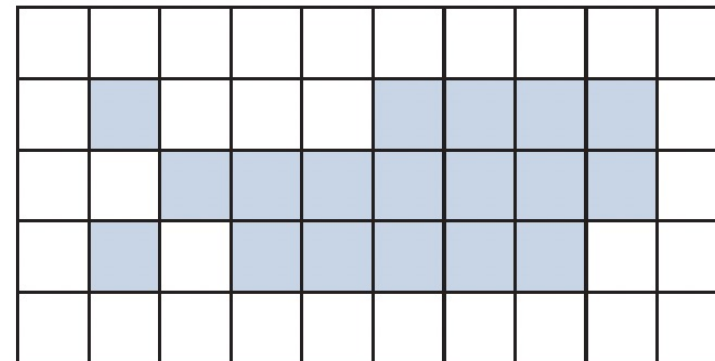
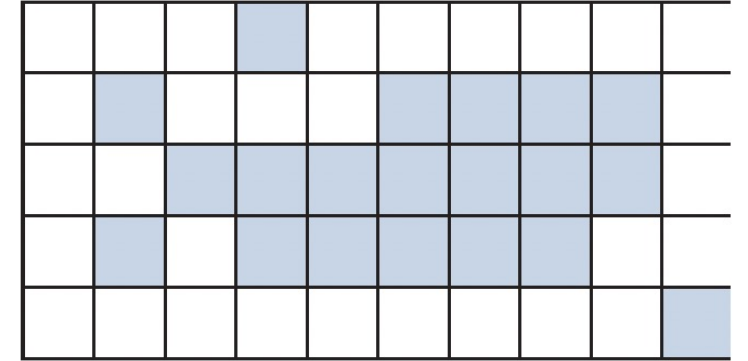
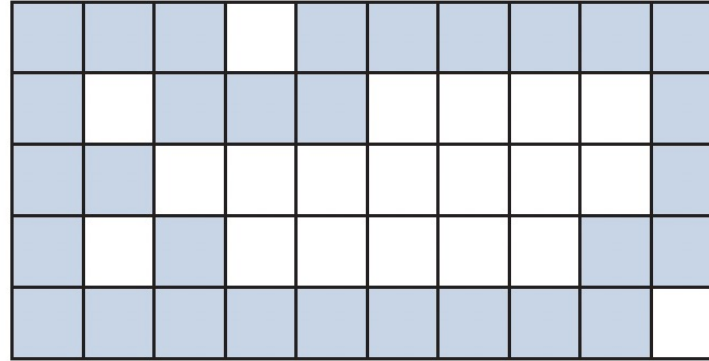
- Dual of Thinning

$$A \odot B = A \cup (A * B)$$

- Note: $A \otimes B = A - (A \odot B)$



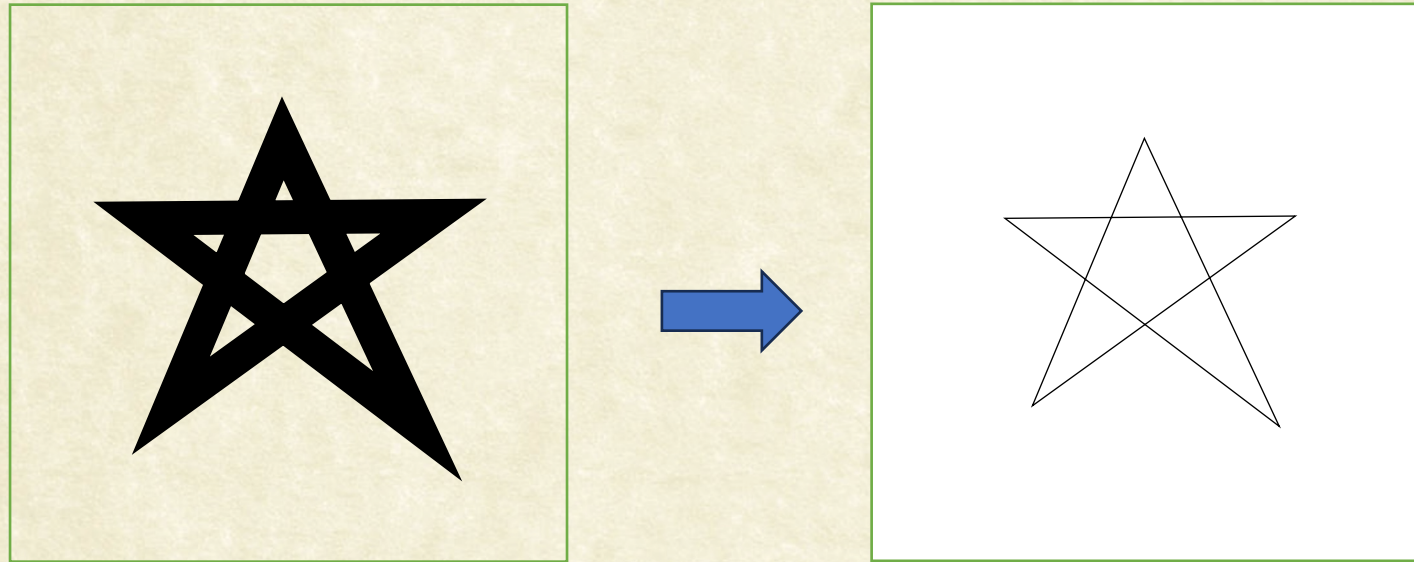
Image, I

 A^c 



CS7.404: Digital Image Processing

Monsoon 2025: Skeletonization

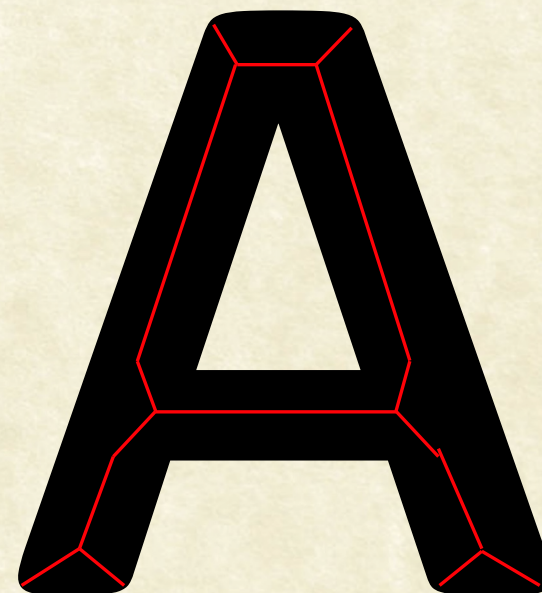


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The Skeleton

- The skeleton of a shape is defined as the set of all points in the shape that are equidistant from the two nearest boundary points.
 - Burning the boundary
- The skeleton of a shape is the locus of the center of maximum disks of the shape





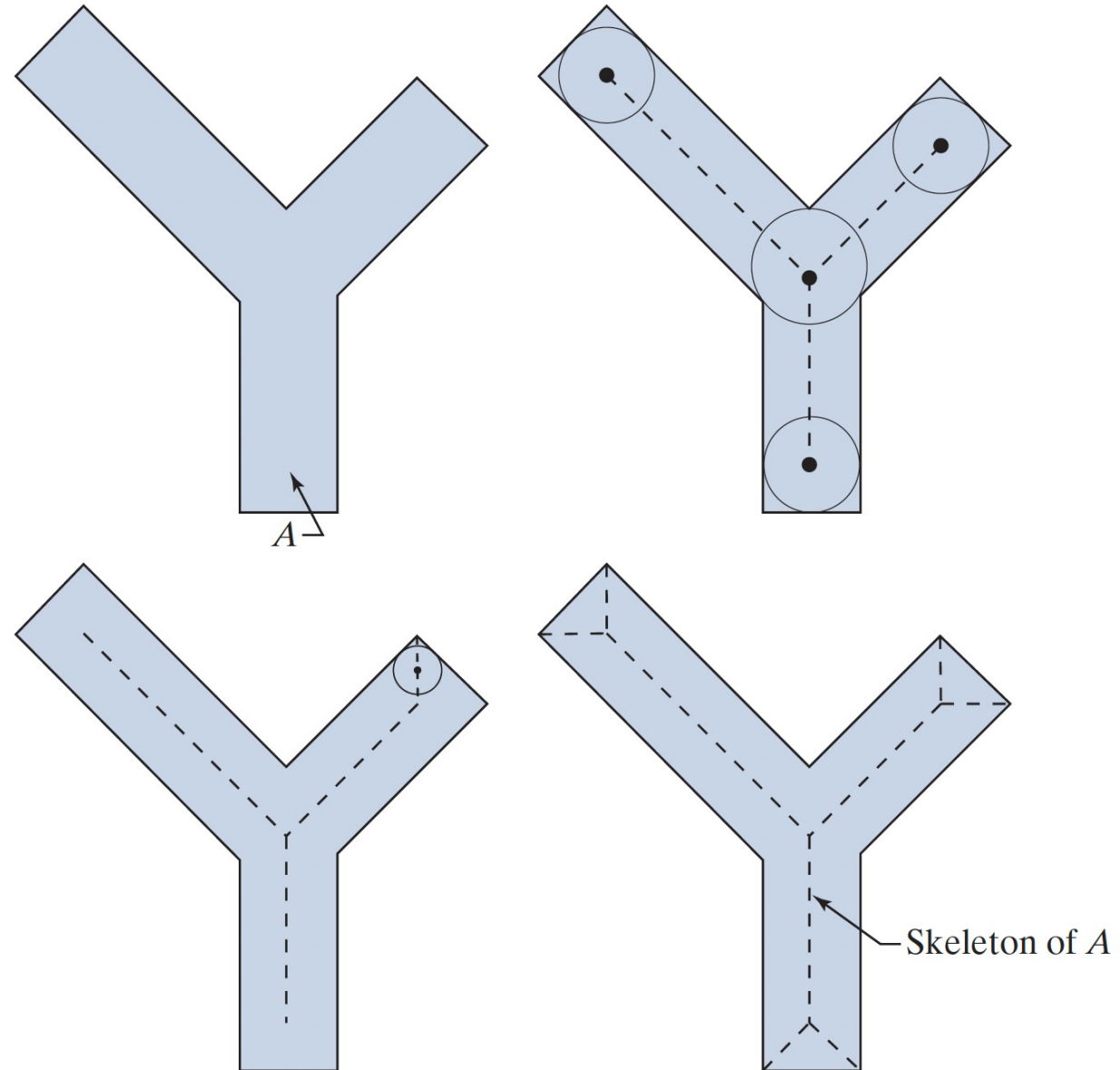
Skeleton as Maximum Disks

- Maximum Disks

- A disc d that fits in A , such that no larger disk exists that fits inside A and contains d .
- A maximum disk will touch the boundary of A at two or more points

- Definition:
$$S(A) = \bigcup_{k=0}^K S_k(A)$$

$$S_k(A) = (A \ominus kB) - (A \ominus kB) \circ B$$





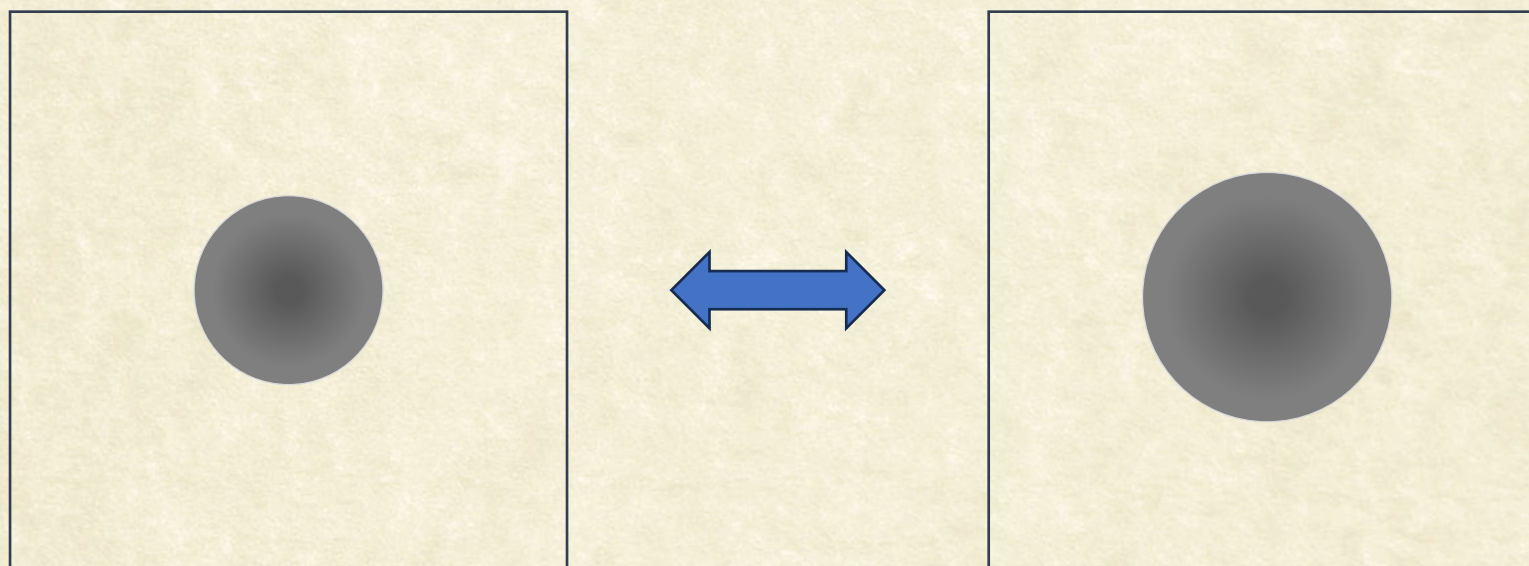
Other Morphological Algorithms

- Hole Filling
- Connected Component Extraction
- Convex Hull
- Pruning
- Border Clearing
- Geodesic Dilation/Erosion
- Morphological Reconstruction



CS7.404: Digital Image Processing

Monsoon 2025: Grayscale Morphology

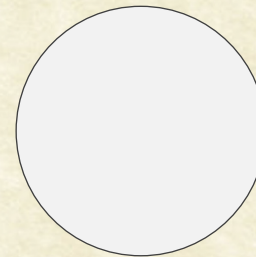
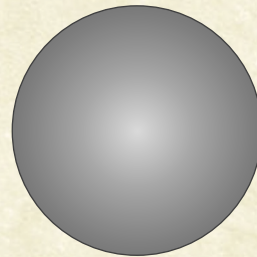


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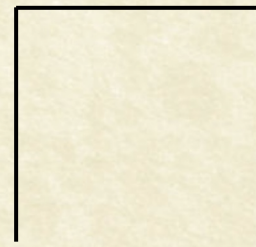
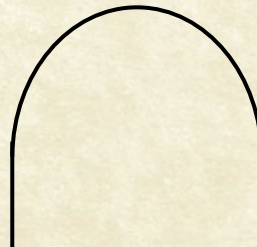


Grayscale Morphology

- Extending the morphological operations to grayscale images.
- Grayscale images, $f(x, y)$ and SEs, $b(x, y)$ are functions that map 2D pixel co-ordinates to its values: $\mathbb{Z}^2 \rightarrow \mathbb{R}$ or $\mathbb{Z}^2 \rightarrow \mathbb{Z}$.
- SEs can be nonflat or flat. Most common are flat, symmetrical SEs of unit height.



Nonflat vs flat SEs



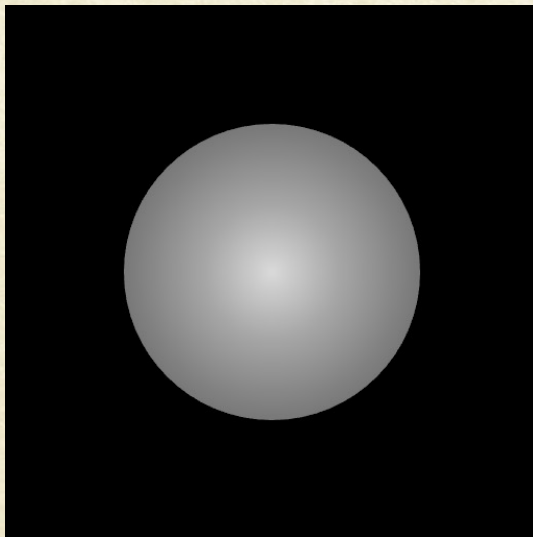
Intensity Profiles



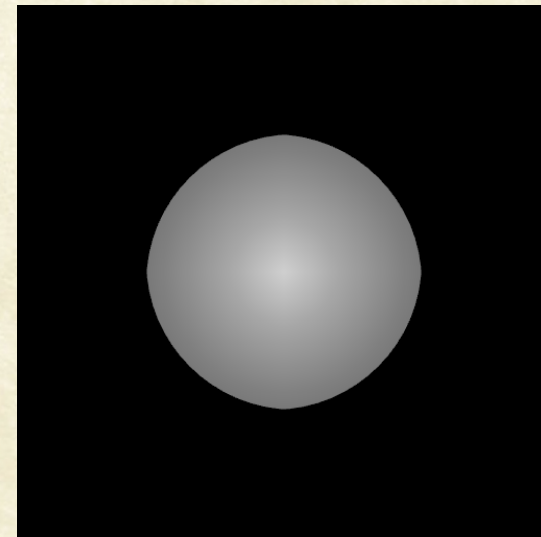
Grayscale Erosion

- Grayscale erosion by a flat SE, b at location (x, y) is defined as *the minimum value* of the image in the region coincident with $b(x, y)$, when the origin of b is at (x, y) .

$$[f \ominus b](x, y) = \min_{(s,t) \in b} \{f(x + s, y + t)\}$$

 $\ominus$

1	1	1
1	1	1
1	1	1

 $=$ 



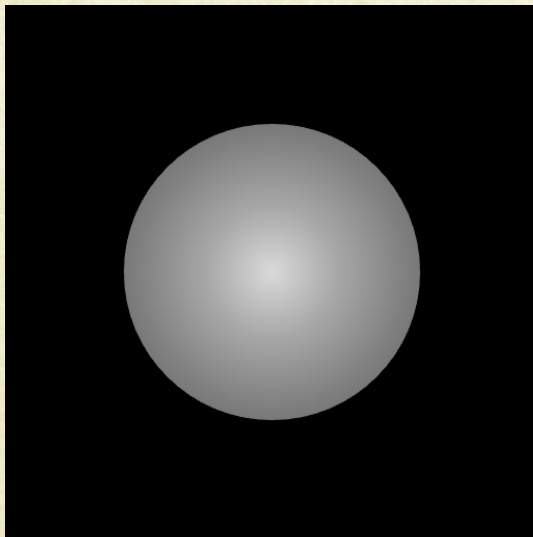
Grayscale Dilation

- Grayscale dilation by a flat SE, b at location (x, y) is defined as *the maximum value* of the image in the region coincident with $\hat{b}(x, y)$, when the origin of $\hat{b}$ is at (x, y) .

$$[f \oplus b](x, y) = \max_{(s, t) \in b} \{f(x - s, y - t)\}$$

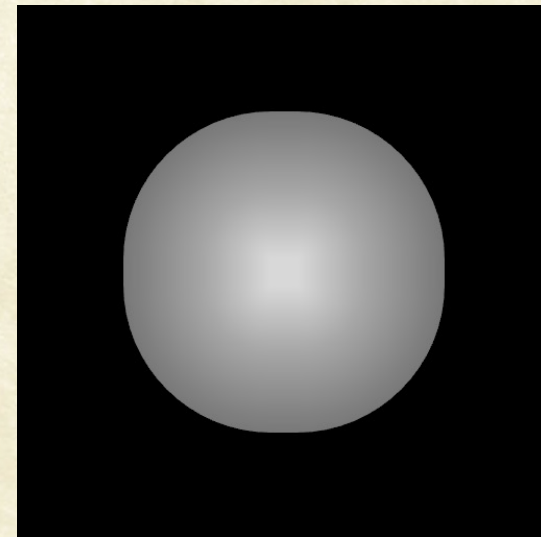
Note:

$$\hat{b}(c, d) = b(-c, -d)$$



1	1	1
1	1	1
1	1	1

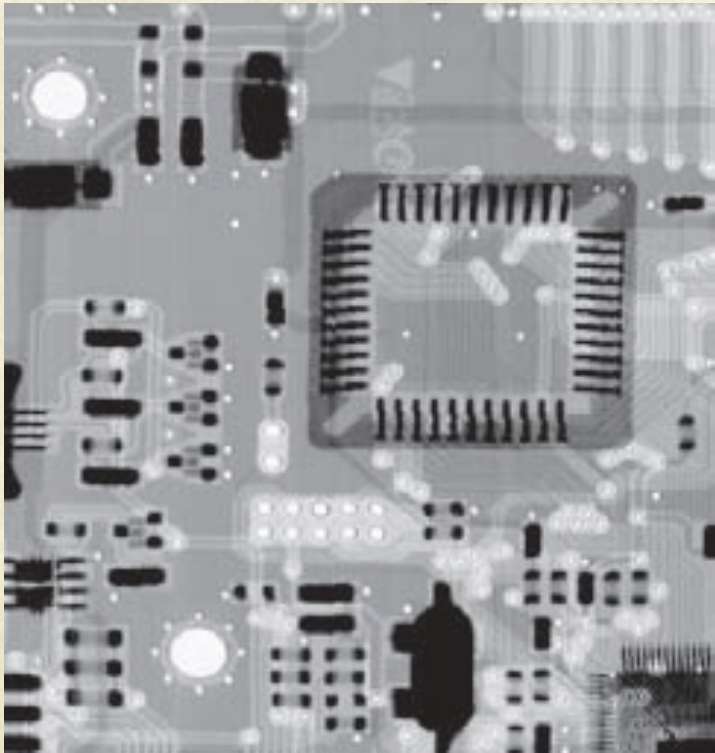
=



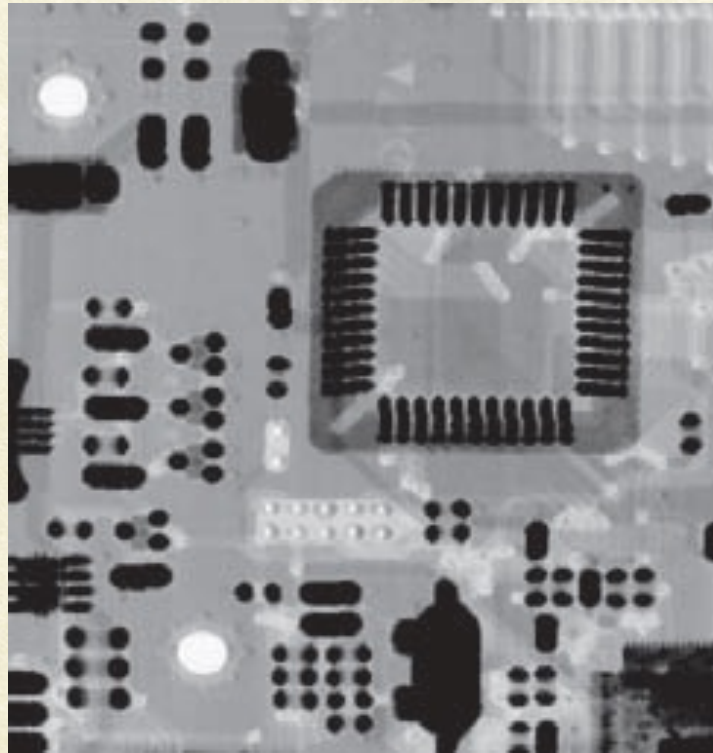


Grayscale Erosion and Dilation: Example

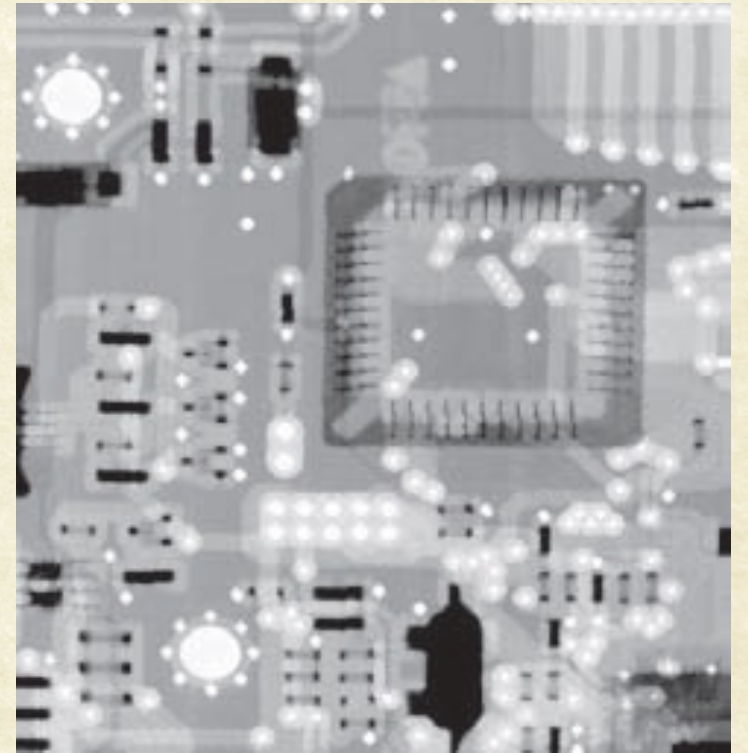
Original



Eroded



Dilated





Grayscale Opening and Closing

Opening: $f \circ b = (f \ominus b) \oplus b$

Closing: $f \cdot b = (f \oplus b) \ominus b$

Duality w.r.t. complements:

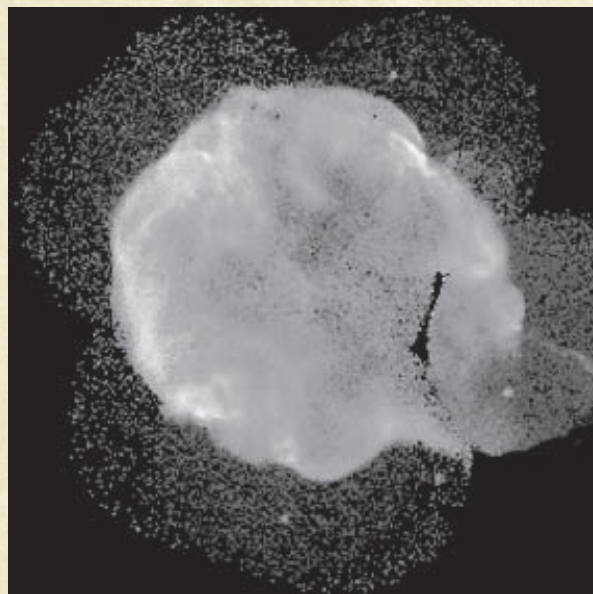
- $(f \circ b)^c = f^c \cdot \hat{b}$
- $(f \cdot b)^c = f^c \circ \hat{b}$
- Question: Are grayscale opening and closing idempotent?



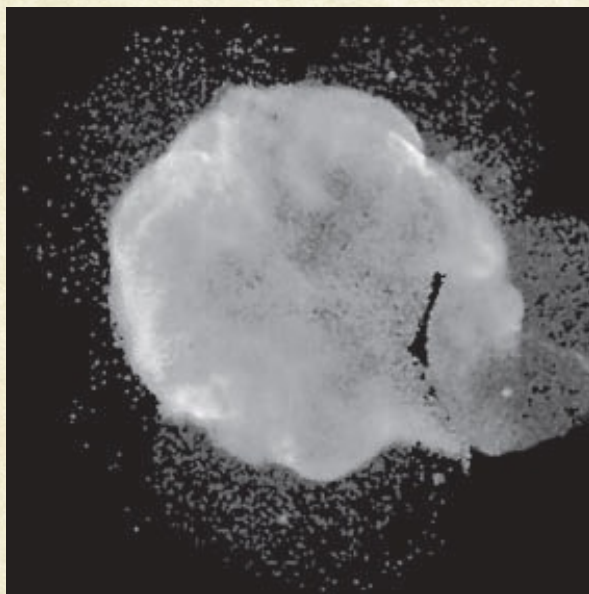
Grayscale Opening: Example

- Opening followed by closing can remove small bright spots.

Cygnus Loop Supernova
in X-Ray band (HST)



Disk radius 1



Disk radius 3



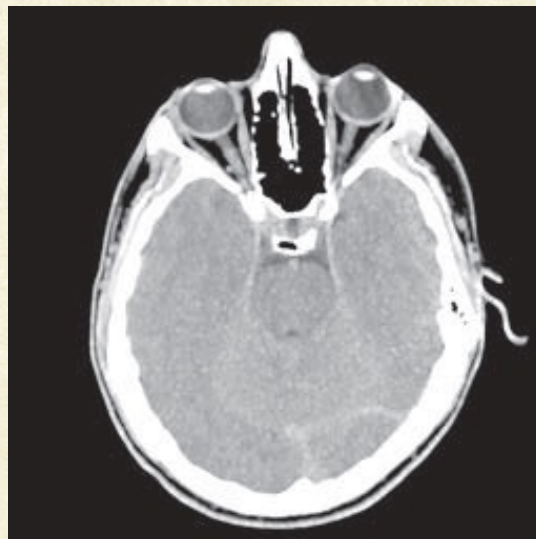
Disk radius 5



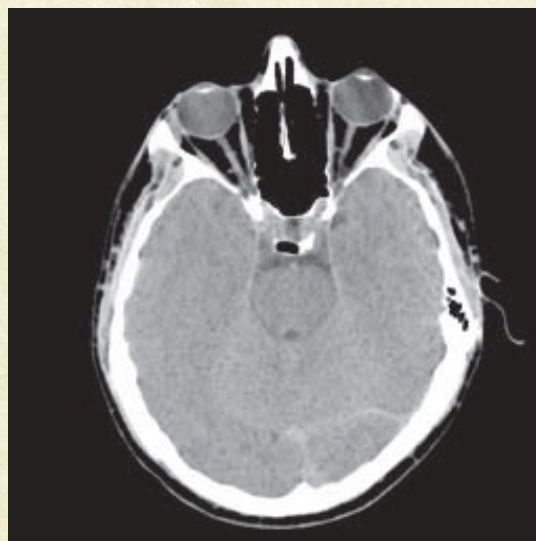


Morphological Gradient: $g = (f \oplus b) - (f \ominus b)$

$(f \oplus b)$



$(f \ominus b)$



Morphological Gradient





Morphological Operations: Summary

- Effective for structural analysis
- Often highly efficient
- Provides fine control through SEs
- Applications:
 - Image Enhancement
 - Pattern Matching
 - Defect Detection
 - Granulometry
 - Shape Analysis ...



Questions?